D&G Pricing Analysis

Executive Summary & Findings

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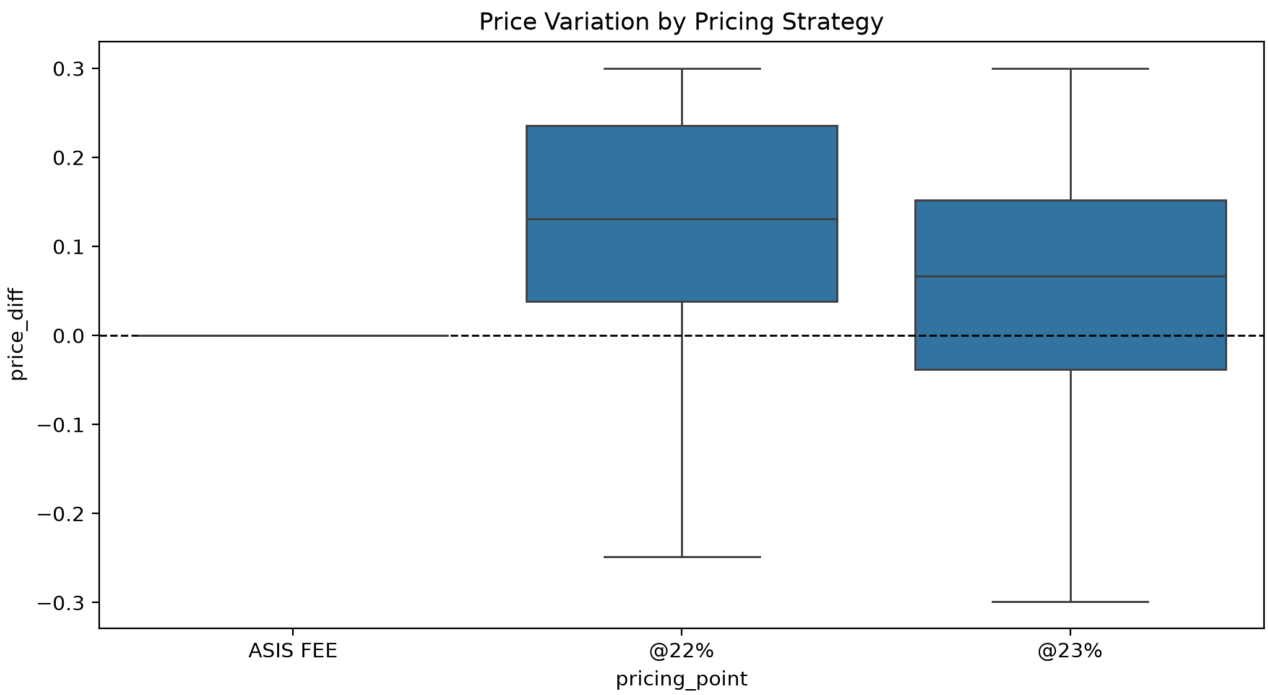
Executive Summary

This report analyses offer-level pricing data across three strategies (ASIS, @22%, @23%) to evaluate conversion, revenue, elasticity, and claims relationships. @22% is recommended as the primary pricing strategy, delivering significantly higher revenue uplift than @23% with no statistically proven cost to conversion. Claims history was found to be an independent, strong driver of conversion that could improve future targeting.

A. Exploratory Data Analysis

Offer volumes and customer/appliance profiles were broadly comparable across strategies. ASIS price_diff is exactly zero (no optimisation applied), while @22% and @23% show a spread of price increases and decreases. predictedconversionrate was found to be a constant placeholder (1.0) for all ASIS rows, excluded from later model calibration checks.



B. Pricing Strategy Performance

	no_of_offers	avg_offered_pre	avg_price_incre	conversion_rate
ASIS FEE	1115.00	48.97	0.00	0.23
@22%	3923.00	55.21	0.12	0.22
@23%	3827.00	51.98	0.06	0.23

Chi-square test found no significant difference in conversion between strategies ($p = 0.61$). Welch's t-tests confirmed offered premium differs significantly between every strategy pair ($p < 0.001$).

C. Price Elasticity

	elasticity
@22% vs ASIS	-0.44
@23% vs ASIS	-0.46
@22% (low vs hi	0.66
@23% (low vs hi	0.17

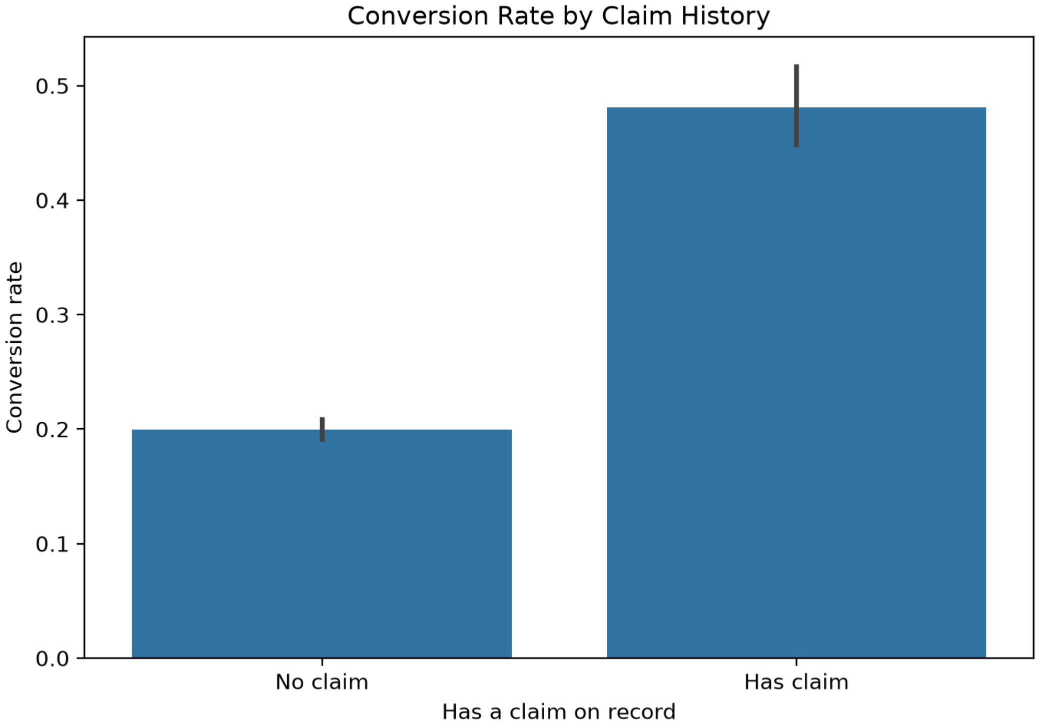
Between-group elasticity was negative as expected (@22%: -0.44, @23%: -0.46). Within-group elasticity was positive for both strategies (@22%: +0.66, @23%: +0.17), traced partly to a selection effect where the optimisation model links price to its own predicted conversion rate for @22% ($r = +0.11$).

D. Strategy Ranking and Model Bias

	best_strategy	value
Statistical: co	@23%	0.23
Statistical: pr	@22%	55.21
Revenue: avg re	@22%	6.42
Revenue: total	@22%	5545.08

@22% outperforms @23% on revenue (avg uplift £6.42 vs £2.92; total uplift £5,545 vs £2,535). Calibration gaps for both optimised models were small (0.008 and 0.004), though @22% shows mild selection bias linked to predicted conversion, worth monitoring if scaled further.

E. Claims and Conversion



Chi-square confirmed a highly significant relationship between claims and conversion (chi2 = 324.81, $p < 0.0001$). Claim holders converted at 48.1% vs 20.0% for non-claimants, roughly 2.4x higher, consistent across all three strategies and independent of plan ownership.

F. Final Recommendation

Adopt @22% as the primary pricing strategy, replacing ASIS, given its clear revenue advantage and lack of statistically proven conversion cost ($p = 0.61$). Continue monitoring @23% as a lower-risk alternative.

Caveats: conversion differences are not statistically significant overall, so this recommendation leans on revenue evidence. @22%'s mild selection bias should be investigated via a proper A/B test before scaling. Claims history is recommended as a new segmentation variable for future pricing models, given its strong, independent effect on conversion.